TEMPO-FL: Temporary Model Escalation and Selective Promotion for Adaptive Hierarchical Federated Learning

Mohammad Matin Rezaeifard
Department of Computer Engineering
Faculty of Engineering
University of Guilan
Rasht, Iran
matinrezaeifard@gmail.com

Reza Paktinat Keleshteri
Department of Computer Engineering
Faculty of Engineering
University of Guilan
Rasht, Iran
reza81paktinat@gmail.com

Fatemeh Zahedi
Department of Computer Engineering
Faculty of Engineering
University of Guilan
Rasht, Iran
dylanzhdi@gmail.com

Reza Ebrahimi Atani
Department of Computer Engineering
Faculty of Engineering
University of Guilan
Rasht, Iran
rebrahimi@guilan.ac.ir

***Abstract*—Federated learning enables collaborative training without sharing raw data, but standard federated averaging produces a single global model that cannot accommodate heterogeneous client data. Personalized and hierarchical approaches improve adaptation and communication, yet lack input-dependent model selection that preserves a client-specific specialist as the default. This paper proposes TEMPO-FL, an adaptive hierarchical federated learning framework that keeps a local specialist model as the base, caches more general models as shadow models, and escalates uncertain inputs based on calibrated confidence. Clients contribute to higher-level models through shadow participation without adopting them for inference. Persistent reliance on higher-level models triggers selective knowledge transfer via replay and distillation, accepted only if local specialization is preserved. Experiments on the TrashNet dataset with eight non-IID clients and one hundred communication rounds compare TEMPO-FL with federated averaging, hierarchical federated learning, and Ditto. TEMPO-FL improves owned-class accuracy and macro-F1 for several clients, with owned-class gains of up to 38.21 percentage points in accuracy and 44.49 points in macro-F1 over their corresponding global-test performance, while retaining the specialist model as the default inference path. These results suggest that temporary escalation and selective promotion can preserve client specialization while exploiting hierarchical knowledge.**

***Index Terms*—federated learning, hierarchical federated learning, knowledge distillation, personalization, selective promotion, uncertainty-aware inference.**

I. Introduction

Federated learning (FL) enables a large number of distributed clients to collaboratively train a model without transferring their raw data to a central server [1]. In each communication round, participating clients perform local training on their own data and transmit only model updates, which the server aggregates into a shared global model. This paradigm has been shown to be robust under non-IID and unbalanced client data distributions while substantially reducing the communication rounds required for convergence [1]. However, standard federated averaging (FedAvg) maintains a single global model that is exchanged between every client and the server. As a result, all clients are guided toward the same shared representation regardless of how different their local data distributions are, and no client receives a dedicated model adapted to its own data. Recent surveys have documented this limitation alongside the broader architectural, synchronization, and security challenges of FL deployments [9], [13].

Statistical heterogeneity is the primary reason why a single global model is often insufficient. Personalization addresses this issue by allowing each client to maintain a model adapted to its own distribution. A common mechanism regularizes the client model toward a jointly trained global model through a tunable interpolation weight, so that the client can rely more heavily on either its local data or the global consensus [2]. Ditto jointly learns a shared global model and personalized client models through global regularization, and has been shown to improve accuracy, fairness across clients, and robustness to poisoned updates simultaneously [2]. In this setting, personalization is performed at the client-model level, and there is no hierarchical structure that allows inputs to be routed selectively to models at different levels of generality. In addition, maintaining a personalized model alongside a global model increases local storage and computation at every client.

A complementary line of work reduces the communication cost of FL by introducing an intermediate aggregation layer between clients and the cloud. In client-edge-cloud architectures, nearby clients are first aggregated at an edge server, and only the resulting edge-level models are periodically syn-

chronized with the cloud, which reduces costly long-distance communication [3], [10]. The relative aggregation frequency between edge servers and the cloud directly affects the trade-off between training speed and device energy consumption [3]. The same principle has been applied to wireless networks, where clustered clients are aggregated by small-cell base stations and periodically reconciled by a macro base station, with gradient sparsification used to reduce latency [4]. Hierarchical federated learning (HFL) therefore reduces communication overhead through multi-level aggregation, but the models at each level remain shared among the corresponding client groups rather than being personalized for individual clients.

Personalization has also been combined with hierarchical aggregation. In personalized hierarchical federated learning, personalized edge-level models are formed by combining locally aggregated and cloud-level models through learned mixing coefficients, complemented by client-side learnable normalization and similarity-based cloud aggregation [5]. While this improves over non-personalized HFL, personalization is achieved by forming an edge-level personalized model through learned model mixing. There is no input-dependent inference mechanism that selectively escalates individual inputs to more general models according to their confidence, and no mechanism that promotes useful higher-level knowledge back into a client-specific model on the basis of sustained evidence.

The above discussion focuses on accuracy, communication, and personalization. A parallel line of work addresses security and privacy concerns in FL. Byzantine-robust aggregation has been studied to tolerate adversarial clients that send corrupted updates [8], and privacy-preserving decentralized training has been explored through randomized pruning and peer exchange [10]. Poisoning attacks against privacy-preserving FL and corresponding defenses have also been catalogued [11], and backdoor-based covert channels have been shown to be embeddable within model updates [12]. Comprehensive surveys of attacks, defenses, frameworks, and applications are available [9], [13]. TEMPO-FL does not provide formal privacy or robustness guarantees, and is designed to be compatible with such mechanisms. The implications of cached shadow models and shadow participation for privacy and robustness are discussed in Section IV.

In summary, the literature leaves a gap at the intersection of personalization, hierarchy, and inference-time adaptivity. FedAvg provides a flat global model without client-specific specialization [1], whereas personalized FL introduces client-level adaptation without hierarchical inference [2]. Hierarchical FL introduces multiple aggregation levels but does not provide client-specific personalization [3], [4], whereas personalized hierarchical FL extends personalization to the edge level but does not provide input-dependent model selection or selective knowledge promotion across the hierarchy [5]. Existing security and privacy mechanisms operate orthogonally to this gap, and none of them addresses inference-time model routing. These limitations motivate a framework in which a client-specific specialist model remains the default inference path, uncertain inputs can be selectively escalated to more general models, and useful higher-level knowledge can subsequently be transferred back to the specialist model based on sustained evidence.

To illustrate the problem, consider a client in a waste-sorting application whose local data primarily contains paper and plastic. Its specialist model performs well on those two classes but is uncertain about a glass image. In a flat FedAvg deployment, the client has no alternative model to consult and must accept an unreliable prediction. In a non-personalized HFL deployment, the client must use a general model for all inputs, which sacrifices specialization on its own classes. TEMPO-FL addresses this trade-off by allowing the specialist to remain the default while uncertain inputs are selectively escalated to more general models, and by transferring useful higher-level knowledge back to the specialist only when sustained evidence supports it.

This setting raises three research questions. Can a client-specific specialist retain owned-class performance while a hierarchical global model achieves comparable or better global performance? Does confidence-based escalation reduce reliance on general models without sacrificing accuracy on uncertain samples? Finally, Does persistent-reliance-driven promotion improve the specialist without catastrophic forgetting of previously acquired specialization? Addressing these questions is non-trivial for several reasons. Calibration across heterogeneous clients requires per-model temperature scaling on data that may not represent the escalation distribution. Distinguishing transient from persistent distribution shift requires a monitoring window that avoids both false positives and slow detection. Knowledge transfer from a higher-level model must preserve local specialization while absorbing new knowledge. Finally, when the hierarchy is updated, the escalation thresholds of the lower-level models must be recalibrated to remain valid.

This paper proposes TEMPO-FL, an adaptive hierarchical federated learning framework built around three distinct processes: federated contribution, temporary utilization of higher-level models, and persistent knowledge transfer from higher-level models. Each client maintains a local specialist model, referred to as the base model, which is optimized on the client's local data and retained as the default inference model. More general intermediate and global models are stored as cached shadow models and are consulted only when the specialist's calibrated confidence falls below a threshold. Clients also participate in hierarchical training through shadow participation, contributing updates to higher-level models without adopting those models as their primary inference path. TEMPO-FL monitors the escalation rate and the resolution rate over time, so that short-term changes in the local data do not trigger a modification of the base model. When increased reliance on higher-level models persists across consecutive time intervals and those models consistently resolve escalated samples, knowledge from the higher-level model is transferred to the base model through a combination of local replay and knowledge distillation [6]. A candidate model is accepted as the new base model only if it preserves local specialization

and improves the ability to resolve samples that previously required escalation. In this manner, TEMPO-FL exploits more general knowledge without automatically eliminating the specialization of the local model.

The remainder of this paper is organized as follows: Section II defines the problem and describes the TEMPO-FL framework. Section III presents the experimental setup and results. Section IV discusses security, privacy, and limitations. Section V concludes the paper.

II. Proposed Idea

In this section, the TEMPO-FL framework is described as an Adaptive Hierarchical Federated Learning approach, which has been designed with the goal of preserving the specialization of local models while flexibly leveraging more general knowledge at higher levels. In this framework, the specialized model of each node is preserved as the default inference model, and only when the specialized model does not have sufficient confidence regarding a sample, more general models enter the decision-making process. At the same time, nodes participate in the hierarchical Federated Learning process by sending model updates and contribute to the formation and improvement of higher-level models.

TEMPO-FL also determines the degree of dependence of the local model on the higher-level general models over time, so that the degree of dependence is gradually reduced or increased based on the need for more persistent knowledge. In the case of continued dependency and confirmation of the superiority of a higher-level model, the knowledge of that higher-level model is transferred to the base model in a controlled manner, thereby improving the capability of the local model without directly replacing the local model with the general model. Thus, federated contribution, temporary utilization of higher-level models, and persistent knowledge transfer from higher-level models constitute three distinct processes within the TEMPO-FL framework.

A. Problem Definition

Federated Learning System. A Federated Learning system consists of a set of distributed nodes, where each node maintains its own local data and participates in training a shared model without directly transferring the local data. The set of clients is denoted by $\mathcal{C} = \{C_1, C_2, \ldots, C_N\}$.

Each client i is assumed to possess a local dataset defined as:

$$D_i = \{(x_j, y_j)\}_{j=1}^{n_i} \tag{1}$$

where $n_i = |D_i|$ represents the number of local samples. In communication round t, a subset of participating clients $S_t \subset C$ is selected, and the model received by each selected client is trained using the corresponding client's local data. The selected clients only update the model parameters and send the updated parameters to the server for aggregation. In the conventional FedAvg method, the aggregated model can be expressed as follows:

$$\theta^{t+1} = \sum_{i \in S_t} \frac{n_i}{\sum_{k \in S_t} n_k} \theta_i^{t+1} \tag{2}$$

Where θ_i^{t+1} represents the parameters of the trained model at client i, and θ^{t+1} represents the shared model parameters after aggregation. Throughout the federated learning process, the raw data of each client remains locally stored at that client, and only model-related information is exchanged during the federated learning process. The principle of retaining raw data locally while exchanging only model-related information is also preserved in TEMPO-FL.

Hierarchical Federated Learning. Assume that the system consists of L levels, and the model at level l is represented by

$$M^{(l)} = f_{\theta^{(l)}} \tag{3}$$

In this definition, f represents the identical network structure at all levels, and $\theta^{(l)}$ denotes the model parameters at level l. Therefore, models at different levels have the same architecture but differ in terms of corresponding model parameters. In the training process, each model receives updates based on the data associated with the corresponding level, meaning that higher-level models have more general knowledge, while lower-level models retain more local specialization. As the hierarchy progresses toward higher levels, the models incorporate broader aggregated knowledge.

Local Specialist Model. Each client i maintains a Local Specialist Model defined as

$$M^{base} = f_{\theta^{base}} \tag{4}$$

The Local Specialist Model is optimized according to the local data distribution of client i and serves as the default inference model. Unlike approaches in which the aggregated model directly replaces the local model after each training stage, TEMPO-FL maintains a distinction between the base local model and more general models. The distinction among these model types aims to preserve personalization, as a single model cannot achieve optimal performance for all clients due to variations in the clients' data distributions.

Cached Shadow Models. In addition to the base model, a client may maintain cached representations of more general models. These models do not replace the base model and are not continuously used for inference. Instead, they remain available as auxiliary models and enter the inference path only when the specialized lower-level model fails to provide a sufficiently reliable prediction.

Uncertainty-Aware Escalation. In TEMPO-FL, the decision-making process begins at the lowest level of the hierarchy. Before inference, the output of each model at level l is independently calibrated using Temperature Scaling [7] so that the resulting confidence score more accurately reflects the actual reliability of the corresponding model. For an input x, the model at level l produces the probability of the sample belonging to each class, and the prediction confidence is defined as

$$C^{(l)}(x) = \max_c p^{(l)}(c \mid x) \tag{5}$$

where $p^{(l)}(c \mid x)$ denotes the calibrated probability assigned to class c by the model at level l. If

$$C^{(l)}(x) \geq \tau_l \tag{6}$$

the prediction produced by the model at level l is accepted. Otherwise, the sample is passed to the next higher level:

$$M^{(l)} \rightarrow M^{(l+1)} \tag{7}$$

The transfer of a sample to a higher level is referred to as Escalation. The purpose of the escalation process is to avoid applying a more general model to all samples and to use broader knowledge only when necessary. If no level is able to provide a sufficiently confident prediction, the sample is treated as Unknown.

Persistent Reliance and Distribution Shift. Temporary escalation to higher-level models does not necessarily indicate that the local model is inadequate. A short-term change in the local data may temporarily increase the need for Shadow Models. Therefore, TEMPO-FL monitors the dependence of each client on higher-level models over time. For client i, the escalation rate to higher levels over time interval t is defined as:

$$R_i(t) = \frac{N_i^{\text{esc}}(t)}{N_i^{\text{total}}(t)} \tag{8}$$

where $N_l^{\text{esc}}(t)$ denotes the number of samples requiring Escalation during time interval t, and $N_i^{\text{total}}(t)$ denotes the total number of samples processed by client i during the same interval. A short-term increase in $R_i(t)$ indicates a transient change and should not result in a modification of the base model. However, if the increased dependence on higher-level models persists across several consecutive time intervals, the persistent dependence may indicate a sustained shift in the local data distribution.

A high level of dependence on higher-level models alone, however, is not sufficient to justify a change to the base model. It is also necessary to determine whether a higher-level model can successfully resolve the escalated samples. The corresponding measure is defined as

$$S_{i,\ell}(t) = \frac{N_{i,\text{resolved}}^{(\ell)}(t)}{N_i^{\text{esc}}(t)} \tag{9}$$

where $S_{i,l}(t)$ represents the proportion of samples escalated to the model at level l that are successfully resolved.

Selective Permanent Promotion. When dependence on higher-level models remains persistent and a more general model demonstrates better performance on the escalated samples, TEMPO-FL evaluates whether the base model should be updated. The base-model update is not performed by directly replacing the local model with the general model. Instead, knowledge from the higher-level model is transferred to the base model in a controlled manner. The knowledge-transfer objective is defined as a combination of local learning and preservation of the knowledge provided by the higher-level model:

$$\mathcal{L} = \mathcal{L}_{\text{local}} + \lambda \mathcal{L}_{KD} \tag{10}$$

where $\mathcal{L}_{\text{local}}$ is used to preserve compatibility with the local data, and $\mathcal{L}_{KD}$ is used to transfer knowledge from the higher-level model. During the promotion process, a candidate model is constructed and accepted as the new base model only if an improvement in performance is confirmed. Therefore, Promotion is a selective and controlled process intended to improve the capability of the local model without eliminating its previously acquired specialization.

Shadow Participation. In TEMPO-FL, a client can contribute to the improvement of more general models at higher levels of the hierarchy without adopting those models as the client's primary inference model. In other words, participation in the training of higher-level models does not imply replacement of the client's local specialist model, and the client's base model is retained for decision-making. During the Shadow Participation process, the client updates a Shadow copy of the higher-level model using the client's local data:

$$\theta_i^{\text{shadow}*} = \text{LocalTrain}\left(\theta^{(\ell)}, D_i\right) \tag{11}$$

Where $\theta^{(l)}$ denotes the model at level l, D_i denotes the local dataset of client i, and $\theta_i^{\text{shadow}*}$ denotes the updated Shadow model after local training. The resulting model update is then used to improve the higher-level model through federated aggregation, while the local specialist model of client i remains unchanged.

B. TEMPO-FL Framework

Figure 1 illustrates the overall TEMPO-FL framework. The objective of the framework is to preserve the specialist model of each client as the primary inference model, while more general hierarchical models are improved through federated learning and are used only when support for local decision-making is required. The TEMPO-FL process begins with the construction of specialist and hierarchical models and continues with adaptive inference, federated participation, and, ultimately, selective knowledge transfer.

In the first stage, an initial general model is provided to the clients as a common starting point, and each client transforms the initial general model into a Local Specialist Model (L1 Specialist) using the client's local data. The Local Specialist Model is retained as the client's permanent base model. The specialist models of the clients are placed at level L1, and aggregation of the L1 models leads to the formation of L2 models. Subsequently, aggregation of the L2 models results in the formation of a more general L3 model. After the hierarchical structure is established, the outputs of the L1, L2, and L3 models are calibrated independently, and a confidence threshold corresponding to the performance of each model is determined.

During inference, each sample is first processed by the L1 specialist model. If the calibrated Confidence of the L1 model exceeds the predefined threshold, the prediction produced by the L1 model is accepted. Otherwise, the sample is escalated to the L2 Shadow model, and the L3 model is used only if the L2 model does not provide sufficient confidence. If none

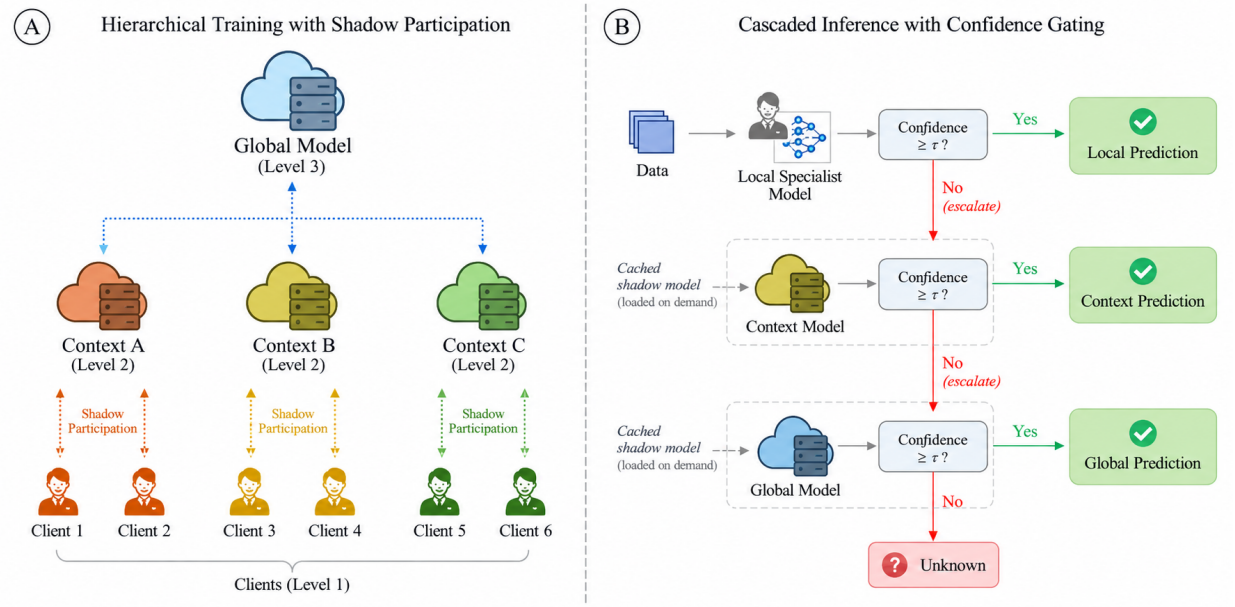


Fig. 1: Overview of the TEMPO-FL framework. (A) Hierarchical organization of the models across three levels: specialized client models at Level 1, context models at Level 2, and the global model at Level 3, with progressively broader knowledge. Clients continuously participate in federated learning through shadow participation. (B) Hierarchical prediction process, where a sample is first evaluated by the specialized model and progressively escalated to the context and global models when verification fails. Samples rejected by all three levels are classified as unknown.

of the available models can provide a reliable prediction, the sample is classified as Unknown. The L2 and L3 models are available as Cached Shadow models for supporting inference. Accordingly, the use of the L2 and L3 models is temporary and sample-specific, and after each decision, the L1 model remains the primary inference path.

Concurrent with inference, the clients participate in the hierarchical federated learning process through Shadow Participation. In each round, a Shadow copy of the L2 model is trained using the local data of the corresponding client, and the resulting updates are aggregated to improve the Context model without directly modifying the L1 base model. The L2 models are subsequently aggregated to update the L3 model. At each level, the acceptance of proposed model updates is controlled by a Trust-Region mechanism so that undesirable updates are rejected. In addition, after the L3 model is updated, a limited amount of knowledge from the L3 model can be transferred back to L2 only when the transfer improves performance and satisfies the defined protective constraints.

TEMPO-FL also monitors the dependence of each client on higher-level models over time. A short-term increase in the number of Escalations does not, by itself, result in a modification of the local model. Instead, the increased dependence must remain persistent across several consecutive time intervals. In parallel, the proportion of escalated samples that higher-level models successfully resolve with sufficient confidence is evaluated. Therefore, the local-model promotion process can be activated only when persistent dependence is accompanied by confirmed utility of a higher-level model.

Under these conditions, the L2 or L3 model does not directly replace the L1 model. Instead, a candidate model is created from the current base model and trained using a combination of local-data Replay and Knowledge Distillation from reliable Teacher samples, following the teacher–student paradigm of knowledge transfer [6]. The candidate model is accepted only if the candidate model preserves overall performance and local specialization while improving the ability to resolve samples that previously required greater reliance on higher-level models. After acceptance, the new model is recalibrated, and the baseline used for dependence monitoring is adjusted according to the updated state. In this manner, TEMPO-FL enables the use of more general knowledge without automatically eliminating the specialization of the local model during federated learning or adaptive inference.

III. Evaluation

A. Experimental setup

Dataset and Models. All experiments were conducted on the TrashNet dataset. The dataset comprises six classes: trash, paper, cardboard, plastic, glass, and metal. Images with an input size of 128×128 pixels are fed into the model. At the beginning of the process, the data are divided into four subsets: 60% for initial training, 10% for validation, 15% for online evaluation, and 15% for testing.

To simulate a non-IID data distribution, each client is assigned one primary class. The assignment is performed implicitly so that the local model of each client specializes

TABLE I: Client structure and class assignments.

Model Hierarchy			Owned Classes
L3	**L2**	**L1**	
Global Model	Education	C00	cardboard, paper
		C01	paper, plastic
	Cafeteria	C02	glass, plastic
		C03	metal, plastic
	Laboratory	C04	glass, metal
		C05	cardboard, glass
	Dormitory	C06	paper, trash
		C07	plastic, trash

mainly in the class assigned to that client. In contrast, higher-level models use data associated with a larger number of clients and, as a result, learn a more general representation of the task.

Federated Learning settings. The federated learning process is simulated using eight clients, denoted by $C_0, C_1, ..., C_7$. In each communication round, the participating clients update their local models using their respective local datasets and send the updated model parameters for aggregation. The training process is carried out for a total of 100 communication rounds.

To evaluate performance, the proposed TEMPO-FL method is compared with three different approaches. The three comparison methods are as follows:

- FedAvg: In this approach, each client updates its local model using the client's local data, and the server constructs a shared global model by performing a weighted average of the local models.
- HFL: In this approach, aggregation is performed at multiple levels. The client models are first aggregated at an intermediate level and are subsequently coordinated at a higher level. This hierarchical structure enables evaluation of the benefits of hierarchical aggregation relative to the standard FedAvg approach.
- Ditto: In addition to a global model, Ditto learns a personalized model for each client and uses regularization to relate the client-specific personalized model to the global model.

All methods were evaluated under the same conditions in terms of the number of clients, local data, communication rounds, and evaluation settings to ensure a fair comparison across the methods. Details of the class assignments, client structure, and model levels are provided in Table I. The table summarizes the evaluation scenario used for TEMPO-FL and the three comparison methods.

B. Results

This section presents the results of the experiments conducted to examine the behavior of the proposed models and to compare TEMPO-FL with the existing methods. The main objective of the evaluation is to determine whether the proposed structure can preserve the specialized client models, improve the performance of the specialized client models on their assigned classes, and effectively utilize more general models at higher levels of the hierarchy. The implementation code and experimental settings are available in the following repository: GitHub Repository [URL].

Table II presents the progress of the TEMPO-FL models at levels L2 and L3 during the federated learning process on the Global Test set. The results in Table II show that, as federated training continues, the higher-level models gradually benefit from the aggregation of client updates, and the performance of the higher-level models improves over the simulation rounds.

The observed trend indicates that federated aggregation enables more general models to gradually accumulate knowledge obtained from different clients. As a result, the L2 and L3 models develop a more general representation of the data during training, while the L1 models can continue to preserve specialization for the client-specific classes. This behavior constitutes a fundamental component of the TEMPO-FL architecture, because the more general models in the hierarchy are intended to provide broader knowledge that can support the more specialized models under conditions of uncertainty.

Table III reports the performance of each client model on two test sets, Global Test and Owned-Class Test. The comparison is intended to characterize the specialized nature of the client models. As the results indicate, a client model does not necessarily achieve uniform performance across all classes represented in the Global Test set; rather, the performance of each client model is primarily associated with the classes present in the corresponding client's local data. For this reason, evaluation on the Owned-Class Test set measures the ability of each client model to address the task for which the model was trained.

Accordingly, the difference between the Global Test and Owned-Class Test results indicates that client-level models can function as specialized models for their respective local data distributions, rather than merely serving as additional copies of a shared global model.

Table IV reports the accuracy of the client models on the Owned-Class Test after 100 communication rounds for the four methods: FedAvg, HFL, Ditto, and TEMPO-FL. The results show that TEMPO-FL achieves higher accuracy on the client-specific classes for several clients. This result suggests that retaining specialized models at the client level can help preserve knowledge associated with each client's local data distribution. In contrast, methods that guide client models more strongly toward a shared representation or global aggregation may lose part of this local specialization.

Table V presents the same evaluation using Macro-F1 as the performance metric. Using Macro-F1 alongside Accuracy provides a more balanced assessment of model performance because Macro-F1 evaluates performance at the class level rather than relying only on the overall number of correct predictions.

The results in Table V follow the same trend observed in Table IV. TEMPO-FL achieves higher performance than the comparison methods on the Owned-Class Test for several clients. The consistency between the Accuracy and Macro-F1 results suggests that the observed improvement is also reflected

TABLE II: Initial and final performance comparison on the global test set.

Shared Generic Baseline: Acc.=55.78%, Macro-F1=53.67%										
Stage	**Global Model**		**Education**		**Cafeteria**		**Laboratory**		**Dormitory**	
	Acc	**Macro-F1**	**Acc**	**Macro-F1**	**Acc**	**Macro-F1**	**Acc**	**Macro-F1**	**Acc**	**Macro-F1**
Initial	63.42	60.26	66.05	62.12	58.42	55.69	54.73	51.40	57.10	55.52
Final	65.78	62.24	66.57	62.99	65.52	62.03	68.15	64.53	61.31	58.71
Final vs. Generic	**+10.00**	**+8.57**	**+10.79**	**+9.32**	**+9.74**	**+8.36**	**+12.37**	**+10.86**	**+5.53**	**+5.04**

TABLE III: Global test and owned-class test performance comparison.

Model	**Global Test**		**Owned-Class Test**		**Owned-Class Gain**	
	Acc	**Macro-F1**	**Acc**	**Macro-F1**	ΔAcc	ΔMacro-F1
C00	61.57	57.49	89.13	89.53	**+27.56**	**+32.04**
C01	61.05	56.53	73.21	79.89	**+12.16**	**+23.36**
C02	60.52	57.82	69.56	76.13	**+9.04**	**+18.31**
C03	57.89	53.97	78.94	76.02	**+21.05**	**+22.05**
C04	54.21	50.91	75.00	76.38	**+20.79**	**+25.47**
C05	53.94	50.41	92.15	94.90	**+38.21**	**+44.49**
C06	60.00	57.49	50.00	54.78	-10.00	-2.71
C07	51.84	50.14	81.48	74.67	**+29.64**	**+24.53**

TABLE IV: Class-wise accuracy comparison of FedAvg, HFL, Ditto, and TEMPO-FL.

Model	**FedAvg [1]**	**HFL [3]**	**Ditto [2]**	**TEMPO-FL**	**Margin**
C00	73.91	73.91	78.26	89.13	**+10.87**
C01	55.36	53.57	55.36	73.21	**+17.85**
C02	47.83	45.65	52.17	69.56	**+17.39**
C03	76.32	78.95	78.95	78.94	-0.01
C04	63.46	59.62	61.54	75.00	**+11.54**
C05	62.75	62.75	60.78	92.15	**+29.40**
C06	55.00	57.50	55.00	50.00	-7.50
C07	59.26	59.26	55.56	81.48	**+22.22**

across the client-specific classes rather than being limited to overall prediction accuracy.

Taken together, these results indicate that the specialized models in TEMPO-FL can better preserve knowledge associated with their local classes and achieve suitable performance on the Owned-Class evaluation. This behavior is consistent with the main objective of the proposed method, namely, maintaining a dedicated specialized model for each client.

To further clarify the structural differences between TEMPO-FL and the comparison methods, Table VI provides a qualitative comparison of the main characteristics of each approach. In Table VI, each method is evaluated according to the presence or absence of several key features of TEMPO-FL.

FedAvg uses a flat architecture with a shared global model and therefore does not provide a hierarchy or client-specific specialization. Ditto provides personalized models for individual clients, but its architecture does not include hierarchical organization or input-level model selection. HFL provides a multi-level structure; however, in its basic form, it does not provide an independent specialized model for each client. pHFL incorporates personalization into a hierarchical

TABLE V: Class-wise Macro-F1 comparison of FedAvg, HFL, Ditto, and TEMPO-FL.

Model	**FedAvg [1]**	**HFL [3]**	**Ditto [2]**	**TEMPO-FL**	**Margin**
C00	83.33	82.37	85.58	89.53	**+3.95**
C01	69.44	67.01	67.62	79.89	**+10.45**
C02	55.42	53.77	56.86	76.13	**+19.27**
C03	83.48	84.57	84.57	76.02	-8.55
C04	68.71	65.19	66.52	76.38	**+7.67**
C05	68.83	68.83	71.45	94.90	**+23.45**
C06	56.42	61.31	56.55	54.78	-6.53
C07	76.36	76.36	74.38	74.67	-1.69

TABLE VI: Comparison of capabilities between TEMPO-FL and existing approaches.

Capability	**[1]**	**[2]**	**[3]**	**[5]**	**TEMPO-FL**
Hierarchical Multi-Level Model Structure	×	×	✓	✓	✓
Client-Level Personalized Model	×	✓	×	×	✓
Confidence-Based Hierarchical Escalation	×	×	×	×	✓
Selective Promotion of Higher-Level Knowledge	×	×	×	×	✓

structure, but personalization is performed at the model/edge level, and the method does not provide an input-dependent mechanism for model selection during inference or selective promotion based on sustained utility.

In contrast, TEMPO-FL integrates these capabilities within a single framework: a client-specific specialized model is retained as the default inference path; uncertain inputs can be escalated, based on confidence, to more general models in the hierarchy; and knowledge from higher-level models is transferred back to the specialized model only when the utility of the higher-level knowledge is observed consistently over time.

IV. Discussion

The evaluation in Section III shows that TEMPO-FL improves owned-class accuracy and macro-F1 for five of the eight clients relative to FedAvg, HFL, and Ditto, while retaining the specialist model as the default inference path. However, the results are not uniformly positive. For C03, TEMPO-FL is essentially tied with the best baseline on accuracy (78.94 versus 78.95) but falls behind by 8.55 points on macro-F1. For C06, TEMPO-FL is 7.50 points behind the best baseline on accuracy and 6.53 points behind on macro-F1. For C07,

TEMPO-FL improves accuracy by 22.22 points but loses 1.69 points on macro-F1. These cases indicate that selective promotion can occasionally shift the specialist away from a class that the baseline handles better. C06 owns paper and trash, and trash is the least represented class in TrashNet. When the specialist for C06 is uncertain about trash, escalation to L2 or L3 may resolve the sample using paper-dominant evidence, and promotion may then reinforce this bias. A per-class promotion criterion, rather than an aggregate validation criterion, would likely reduce this effect.

The evaluation has three further limitations. First, all experiments are conducted on a single dataset (TrashNet) with a modest number of clients and communication rounds. Second, the closest related work, ESPerHFL [5], is not included as a baseline, so the comparison against the state of the art in personalized hierarchical FL is indirect. Third, the number of escalations, the number of promoted models, and the communication and computation overhead of cached shadow models are not reported. These omissions limit the strength of the empirical claims.

TEMPO-FL does not provide formal privacy or robustness guarantees. Three aspects are worth noting. First, cached shadow models increase the local memory footprint and, if the server observes escalation queries, may leak information about the local data distribution through the frequency and identity of escalated classes. Second, shadow participation uploads model updates to higher levels, which are subject to the same inference risks as standard FL updates. Third, the trust-region mechanism rejects updates that fall outside a predefined radius, but it has not been evaluated against poisoning or Byzantine attacks. TEMPO-FL is compatible with Byzantine-robust aggregation [8], secure aggregation, and differential privacy, and integrating these mechanisms is a natural extension. The privacy implications of escalation queries and shadow updates, and the robustness of promotion under adversarial clients, remain open questions.

V. Conclusion

This paper proposed TEMPO-FL, an adaptive hierarchical federated learning framework that separates three processes: federated contribution, temporary utilization of higher-level models, and persistent knowledge transfer. Each client retains a local specialist model as the default inference path, caches more general models as shadow models, and escalates uncertain inputs based on calibrated confidence. Clients contribute to higher-level models through shadow participation without adopting them for inference, and knowledge from higher-level models is transferred to the base model only when persistent reliance is accompanied by consistent utility. Experiments on TrashNet with eight non-IID clients and one hundred communication rounds show owned-class accuracy gains of up to 38.21 percentage points and macro-F1 gains of up to 44.49 points over the corresponding global-test performance, with three clients not improving over the best baseline.

Future work will proceed in four directions. First, the evaluation will be extended to additional datasets, a larger number of clients, and a direct comparison with ESPerHFL [5]. Second, the escalation thresholds, the persistent-reliance window, and the promotion acceptance criterion will be formalized and analyzed, ideally with a convergence guarantee for the hierarchical training. Third, the overhead of cached shadow models in terms of memory, computation, and communication will be quantified. Fourth, the privacy and robustness of escalation queries and shadow updates will be studied, with the goal of integrating secure aggregation [8] and differential privacy into the TEMPO-FL pipeline without compromising the specialist default.

References

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," in *Proc. 20th Int. Conf. Artif. Intell. Stat. (AISTATS)*, PMLR, vol. 54, pp. 1273–1282, Apr. 2017.

[2] T. Li, S. Hu, A. Beirami, and V. Smith, "Ditto: Fair and Robust Federated Learning Through Personalization," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, PMLR, vol. 139, pp. 6357–6368, Jul. 2021.

[3] L. Liu, J. Zhang, S. H. Song, and K. B. Letaief, "Client-Edge-Cloud Hierarchical Federated Learning," in *Proc. IEEE Int. Conf. Commun. (ICC)*, pp. 1–6, Jun. 2020, doi: 10.1109/ICC40277.2020.9148862.

[4] M. S. H. Abad, E. Ozfatura, D. Gündüz, and Ö. Erçetin, "Hierarchical Federated Learning Across Heterogeneous Cellular Networks," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, pp. 8866–8870, May 2020, doi: 10.1109/ICASSP40776.2020.9054634.

[5] C. Ma, X. Li, B. Huang, G. Li, and F. Li, "Personalized Client-Edge-Cloud Hierarchical Federated Learning in Mobile Edge Computing," *J. Cloud Comput.*, vol. 13, no. 1, Art. no. 161, pp. 1–17, 2024, doi: 10.1186/s13677-024-00721-w.

[6] G. Hinton, O. Vinyals, and J. Dean, "Distilling the Knowledge in a Neural Network," in *Proc. NIPS Deep Learning and Representation Learning Workshop*, 2015.

[7] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, PMLR, vol. 70, pp. 1321–1330, Aug. 2017.

[8] S. S. Razavi, S. Arsalan Vasegh Rahim Parvar, S. D. Pakdeh, M. Matin Rezaeifard, M. M. Chafi and R. Ebrahimi Atani, "LightFedSelect: A Lightweight Framework for Byzantine-Robust Federated Learning," 15th International Conference on Computer and Knowledge Engineering (ICCKE), 2025, pp. 1-7, doi: 10.1109/ICCKE68588.2025.11273877.

[9] Daniel M. Jimenez-Gutierrez, Yelizaveta Falkouskaya, José L. Hernandez-Ramos, Aris Anagnostopoulos, Ioannis Chatzigiannakis, Andrea Vitaletti,On the security and privacy of federated learning: A survey with attacks, defenses, frameworks, applications, and future directions, Information Fusion, Volume 131, 2026, https://doi.org/10.1016/j.inffus.2026.104155.

[10] S. Ameri and R. Ebrahimi Atani, "FedPrEx: Decentralized Federated Learning with Randomized Pruning and Peer Exchange for Privacy-Preserving Healthcare AI," Computational Sciences and Engineering, (2026): -, doi: 10.22124/cse.2026.33874.1193.

[11] Santanu Basak, Kakali Chatterjee, Robust security enhancements approach for privacy-preserving federated learning under poisoning attacks, Journal of Information Security and Applications, Volume 102, 2026, https://doi.org/10.1016/j.jisa.2026.104612.

[12] M. M. Rezaeifard, F. Zahedi, S. A. Vasegh Rahim Parvar and R. Ebrahimi Atani, "FedCloak: Backdoor-Based Covert Channels in Federated Learning," 16th International Conference on Information and Knowledge Technology (IKT), Tehran, 2025, pp. 292-298, doi: 10.1109/IKT69652.2025.11626789.

[13] Mahmud F, Mahmud F, Rahman RM. A Survey of Federated Learning: Advances in Architecture, Synchronization, and Security Threats. Computers, Materials & Continua, 2026, 86(3): 1. https://doi.org/10.32604/cmc.2025.073519.